

ASSIGNMENT #4

BINF*6210

**IMPACT OF RNA-SEQ NORMALIZATION
CHOICE ON DIFFERENTIAL EXPRESSION:
A CASE STUDY**

Previous large-scale work by Tong et al. systematically evaluated 278 RNA-seq pipelines and showed that pipeline components such as mapping, quantification and normalization jointly affect accuracy, precision and reliability of gene expression, and that these differences carry through to disease-outcome prediction. In this project, my intentions were to implement a smaller-scale comparison focused on differential expression methods and normalization to see how analysis choices change the set of genes considered significant and the biological conclusions drawn.

I chose to compare 3 normalization methods on edgeR using the same RNA-seq dataset to see how variation in normalization could affect the list of significant genes and biological conclusions.

CHOSEN DATASET: AIRWAY

- Gene-level RNA-seq counts from the Bioconductor airway package.
- The package contains a subset of the experiment published by Himes et al. (2014) on glucocorticoid responses in human airway smooth muscle cells

RESEARCH QUESTION

How robust are differential expression results in the airway dataset to the choice of normalization method in edgeR (TMM vs simple library scaling vs UQ)

EXPLORING THE DATASET USED

- **Data structure**

- The airway object is a SummarizedExperiment containing:
 - An assay matrix assay with raw read counts
 - rows: 63677 genes
 - columns: 8 RNA-seq libraries (samples)
 - A sample-metadata table colData (airway) with information about each library such as the treatment, cell line, sequencing run.

- **Key variables for this analysis**

- Response variable: gene-level read counts (per gene × sample).
- Main experimental factor:
 - dex – treatment with dexamethasone, a synthetic glucocorticoid with anti-inflammatory effects that is commonly used to treat asthma
 - "untrt" = untreated control
 - "trt" = dexamethasone-treated
- In my analysis, dex is encoded as a factor group and used to define the design matrix for differential expression.
- Biologically, each treated sample is paired with an untreated sample from the same donor.

- **Groups being compared**

- Two groups: 4 untreated samples vs 4 dexamethasone-treated samples.
- Differential expression analysis is used to identify genes whose expression changes in response to dexamethasone treatment in human airway smooth muscle cells.

Investigating library sizes, i.e. total reads per sample, moderate variation prior to normalization was found

METHODOLOGICAL PIPELINE

Load data



Filter lowly expressed genes



Application of 3 normalization methods

TMM

none

UQ



for each



Compute logCPM



Fit edgeR quasi-likelihood GLM



Identify DE genes (FDR < 0.05)



Compare methods!

DE overlap, Jaccard index

Distribution of logFCs

CoV and within-group correlation

Computational time

NORMALIZATION METHODS AND EVALUATION CRITERIA

Normalization methods (edgeR calcNormFactors)

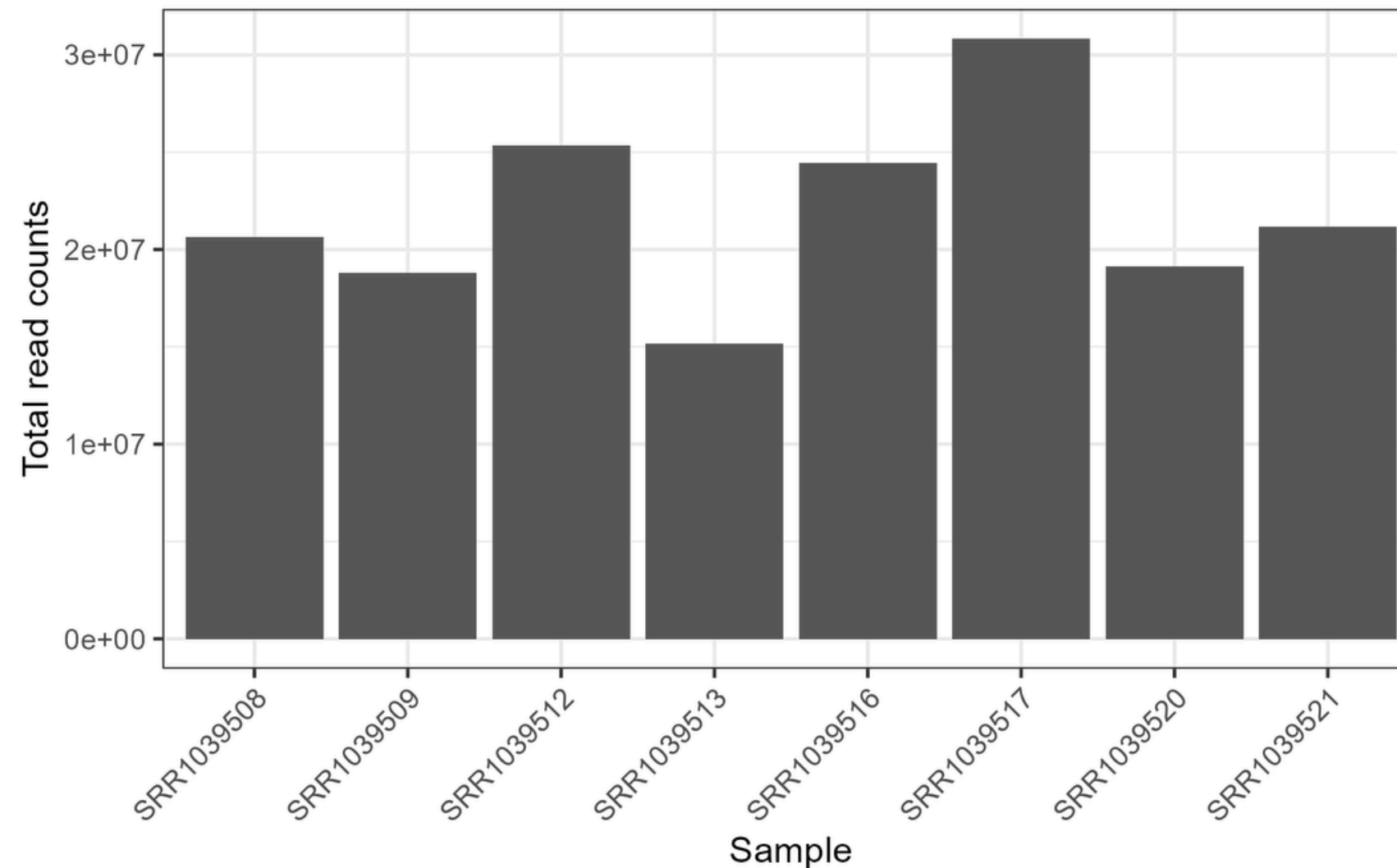
- Simple: normalization method = "none" → this uses only library size scaling (norm.factors = 1).
- TMM (Trimmed Mean of M-values): robust scaling that down-weights genes with large fold-changes or very high counts.
- Upper-quartile (UQ): scales samples so that the 75th percentile of counts (excluding zeros) is equal across libraries.

Evaluation criteria (adapted from Tong et al. 2020)

- Number of DE genes at $FDR < 0.05$ per method.
- Overlap between DE sets (Venn diagram, Jaccard index).
- Precision proxy:
 - Coefficient of variation (CoV) of log-CPM within each group; lower median CoV = higher precision.
- Reliability proxy:
 - Mean pairwise Pearson correlation of normalized log-CPM among biological replicates within each group.
- Computational cost:
 - Run-time for each normalization + DE pipeline (using system.time).

LIBRARY SIZES VARY ACROSS SAMPLES, MOTIVATING BETWEEN-SAMPLE NORMALIZATION BEFORE COMPARING GENE EXPRESSION

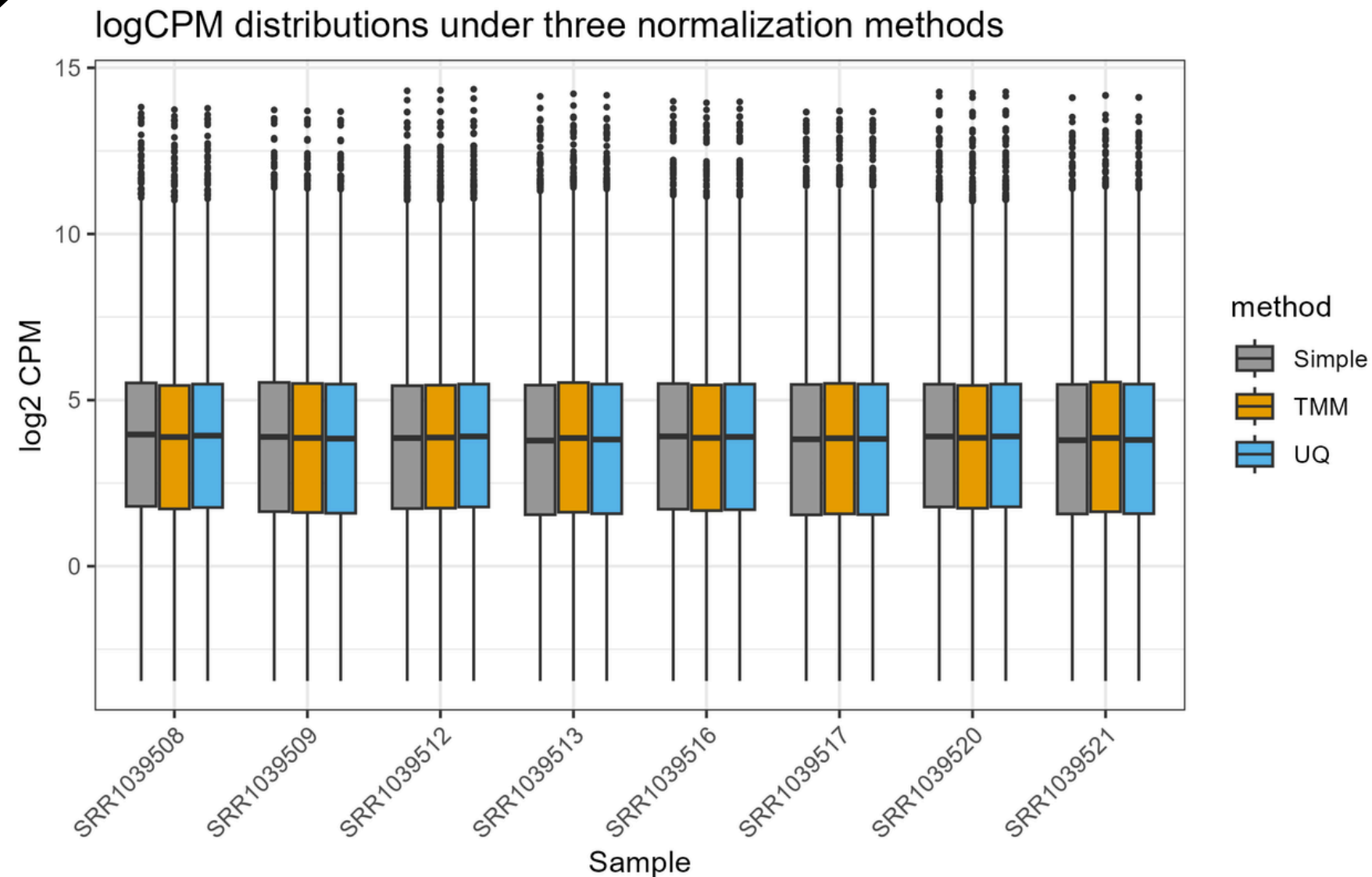
Figure 1: Library sizes per sample



This figure shows total read counts per sample. There is roughly a 2x difference between the smallest and largest libraries, so some normalization is necessary to make samples comparable

FIG: BAR PLOT DEPICTING LIBRARY SIZES PER SAMPLE, WITH COUNTS ON THE Y-AXIS AND SPECIFIC GENES ON THE X-AXIS

ALL THREE NORMALIZATION METHODS PRODUCE VERY SIMILAR LOG-CPM DISTRIBUTIONS



- *The three methods align the medians and overall distributions quite closely.*
- *TMM and UQ give nearly indistinguishable distributions.*
- *Even simple scaling performs well on this balanced, relatively homogeneous dataset*
- *This highlights that library-size differences are modest and composition biases are limited in this dataset.*

FIG: THREE BOXPLOTS (SIMPLE, TMM, UQ) OF LOG2-CPM AFTER FILTERING AND NORMALIZATION FOR EACH SAMPLE

LOG₂ FOLD-CHANGES ARE ALMOST IDENTICAL UNDER TMM AND SIMPLE SCALING, EVEN FOR STRONGLY D.E. GENES

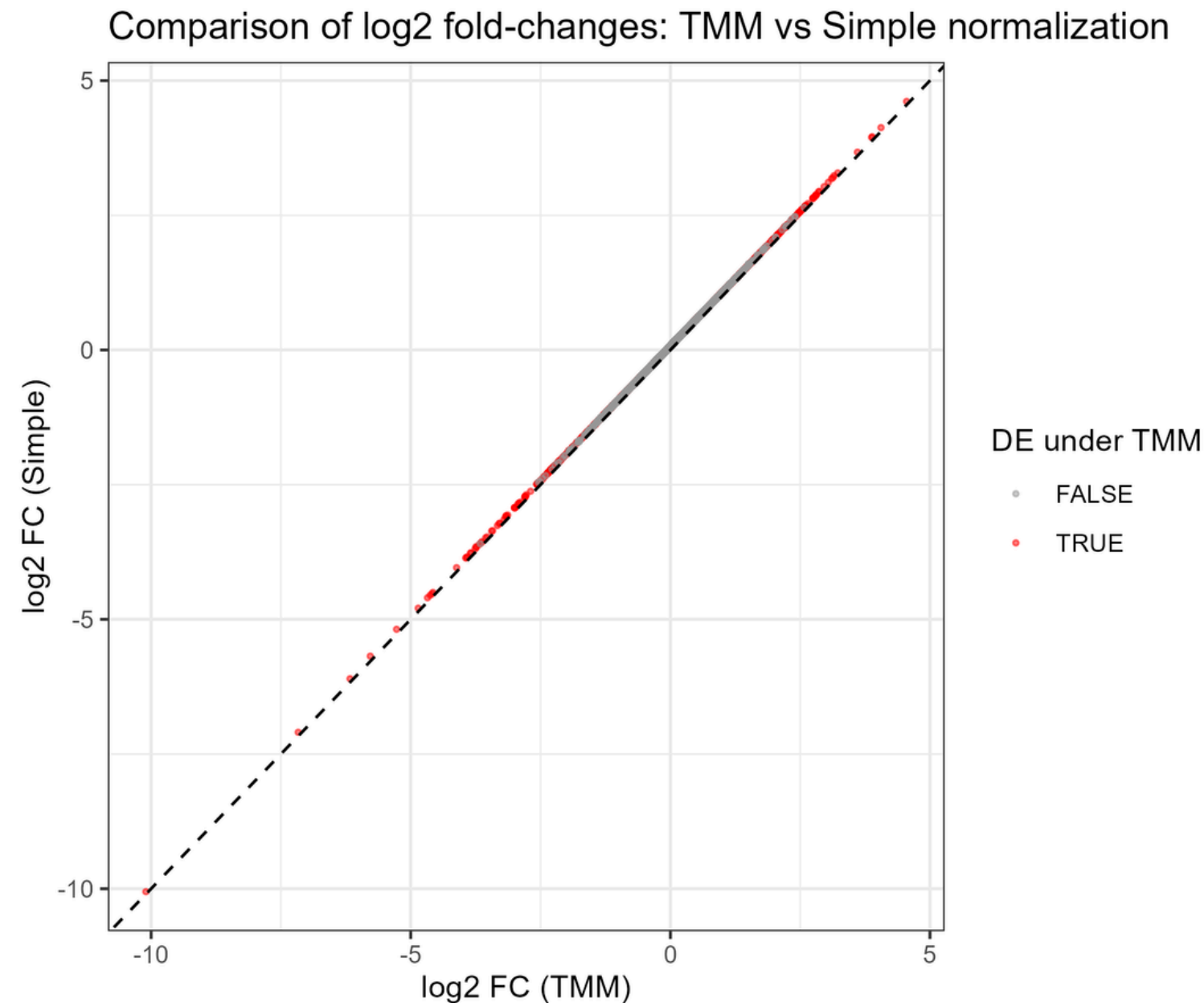
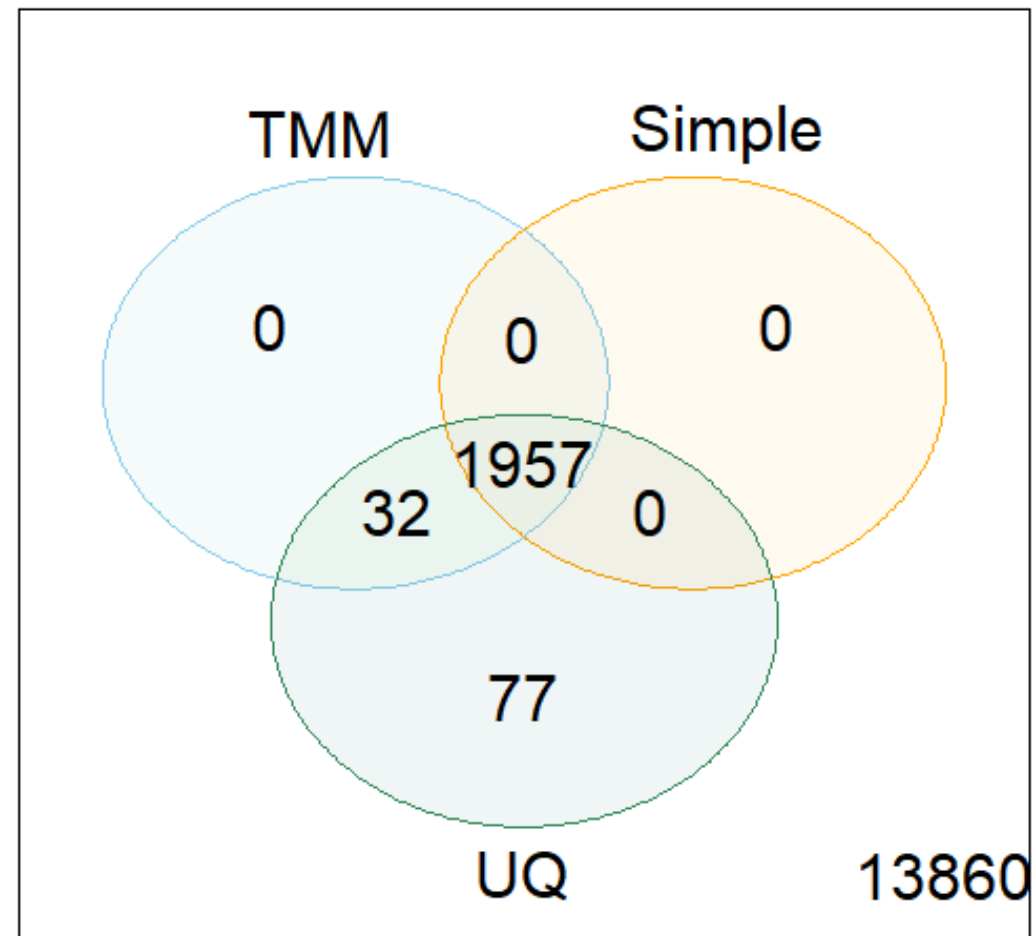


FIG: LOGFC SCATTERPLOT DEPICTING FOR EACH GENE, LOG₂ FOLD-CHANGE UNDER TMM (X-AXIS) VS SIMPLE (Y-AXIS); D.E. GENES UNDER TMM (FDR < 0.05) HIGHLIGHTED IN RED.

- *Almost all points lie on the $y = x$ dashed line, indicating nearly identical log₂FC estimates.*
- *DE genes (depicted in red) follow the same pattern, suggesting that the choice between TMM and simple scaling has minimal impact on effect sizes for this dataset*
- *This implies negligible impact of the normalization choices made on effect size estimates*
- *The logFC scatterplot for TMM vs UQ yielded similar results.*

THE MAJORITY OF DE GENES ARE SHARED ACROSS ALL THREE METHODS, WITH ONLY SMALL METHOD-SPECIFIC SETS

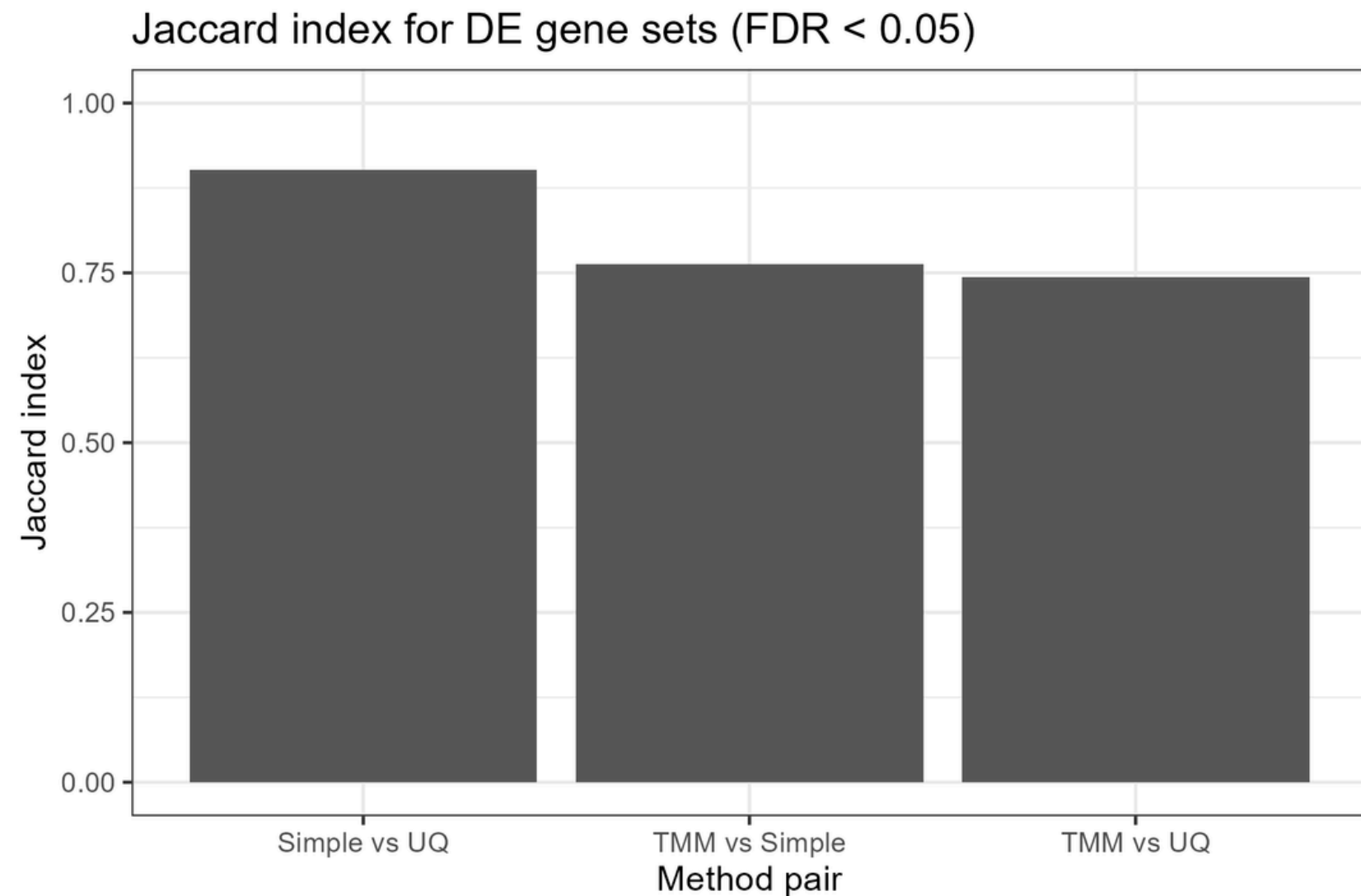
Overlap of DE genes (FDR < 0.05)



- All three methods share a large subset of 1,957 DE genes.
- TMM and UQ identify small additional sets (32 and 77 genes) not detected when there is simple scaling.
- Simple scaling does not contribute unique DE genes in this dataset.
- Suggesting high robustness of DE calls to normalization choice

FIG: VENN DIAGRAM DEPICTING OVERLAP OF D.E. GENES (FDR < 0.05) ACROSS THE THREE PIPELINES

HIGH JACCARD INDICES REINFORCE THAT DIFFERENT NORMALIZATION STRATEGIES PRODUCE HIGHLY OVERLAPPING SETS OF D.E. GENES IN THIS EXPERIMENT



- *Jaccard values are high (>0.7 for all pairs), highest between Simple and UQ.*
- *Confirms quantitatively that DE sets overlap strongly across normalization methods*

FIG: BARPLOT DEPICTING JACCARD SIMILARITY BETWEEN PAIRS OF D.E. GENE SETS

How robust are differential expression results in the airway dataset to the choice of normalization method in edgeR (TMM vs simple library scaling vs UQ)

WHAT WERE THE KEY FINDINGS ?

- Library sizes differed but not dramatically.
- All three normalization methods produced nearly identical logCPM distributions.
- Numbers of DE genes were very similar, with high overlap

DISCUSSING THE RESULTS

- On the airway dataset, simple library-size scaling, TMM and UQ normalization all yield very similar log-CPM distributions, sample clustering, and log2 fold-changes.
- The number of DE genes at $FDR < 0.05$ is large for all methods, and most DE genes (1,957) are shared across all three pipelines.
- Jaccard indices indicate strong overlap in DE calls between methods.
- Thus, in this particular experiment, differential expression results appear robust to reasonable choices of normalization method.

EVALUTION METRICS

- Very similar DE sets across methods suggest that treatment-induced changes are so strong that all three normalizations detect them.
- Precision proxy (CoV of log-CPM):
 - Median CoV within each group differs little between methods, indicating similar within-group variability.
- Reliability proxy (within-group correlations):
 - Mean pairwise Pearson correlations among replicates are high (close to 1) and comparable for all three methods, meaning sample-to-sample expression patterns are stable.
- Computation time:
 - All three pipelines run quickly on this small dataset, and normalization choice seems to have negligible computational impact.

EXPECTED FINDINGS VS ACTUAL FINDINGS

- With the Tong et. al paper serving as the foundational basis to this analysis, it was expected that normalization methods varying would bring about a difference in results, since a key finding from this paper outlined the same. It was an interesting pivot to investigate why this was not the case.

CAVEATS TO ANALYSIS

- The analysis was based on a data of a small sample size ($n = 8$), whose design was fairly balanced, with no extreme library sizes or compositional biases; this may be why simple scaling already performs well. Robinson & Oshlack (2010) show that TMM is particularly helpful when a small subset of genes changes dramatically or when libraries differ strongly in composition.
- Due to a lack of data supporting external ground truth, like the qPCR validation that Tong et.al had, to evaluate absolute accuracy of fold-changes, agreement between methods was used as an indirect proxy. Precision and reliability as well, were evaluated indirectly.
- Computational time was evaluated but noting the small sample size, differences in computational speed were minute. This is likely because runtime is a simple computational performance metric that may vary and hold more weight on larger studies

FUTURE DIRECTIONS

1. Methodologically, following the intended exploration of normalization techniques, it would be pertinent to evaluate the pipeline on different datasets, adapting it to more complex frameworks. Concrete next steps could be to:
 - a. Apply the same normalization comparison to a larger, more heterogeneous RNA-seq dataset where the number of genes are expected to change globally between samples to see if method-specific differences become larger.
 - b. Explore downstream steps highlighted by Tong et al. (2020) to assess their impact on accuracy and reliability in more direct manners, this could be facilitated by incorporating an external benchmark, such as qPCR measurements for a subset of genes.
 - c. Extend analysis to investigating whether normalization choice affects biological interpretation at the pathway level.
2. Conceptually, branching into the biological basis of glucocorticoid treatments on various cell types, including investigating its applications in a veterinary population, would be of personal interest to me.

REFLECTING ON MY EXPERIENCE!

- I chose this area of study because it intrigued me immediately, and I feel emboldened to continue onwards exploring it much further now! Despite being a steep learning curve, each new Bioconductor package, and each research rabbit hole I found myself pursuing, left me with a newfound appreciation for the field of bioinformatics. I consider this project, and the learnings of the entire semester that I built upon to be monumentally instrumental in engaging my intellect in exactly the manner I wanted, going into this program. It makes me excited looking ahead to the next semester.
- Specifically relating to the scope of this project, it was a key lesson learned that on some well-behaved datasets, different “reasonable” choices may give almost identical results, which highlighted the importance of checking robustness rather than assuming sensitivity.
- Next, I would like to explore more complex designs and additional methods and to gain more practice working with different sequencing pipelines. A natural extension of my interest in this area of analysis and my broader interests in veterinary science, would be to consider more relevant datasets that lie in the intersection of the two fields.

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